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Intraluminal Geometry, Compliance, and Contact Measurement Device Vibration, Haptic, and Active Mechanical Stimulus Integration

Field

This disclosure relates to intraluminal measurement devices and, more particularly, to the integration of vibration, haptic, and active mechanical stimulus systems within an insertable device configured to assess internal distance, geometry, compliance, contact, and tissue response within vaginal or anal lumens.

Overview

In certain embodiments, the intraluminal geometry and compliance measurement device further incorporates one or more vibration, haptic, or active mechanical actuation elements. These elements may operate independently or in coordination with internal sensing subsystems to deliver controlled internal tactile stimulus, arousal activation, neuromodulatory input, perceptual feedback, or dynamic mechanical engagement during measurement or assessment.

The device may be configured for vaginal or anal insertion and is designed to be anatomically agnostic and inclusive of multiple use cases.

Actuator Technologies

The vibration or haptic subsystem may include one or more of the following actuator types, individually or in combination:

- Circumferential vibration rings or segmented vibration bands
- Miniature Linear Resonant Actuators (LRA)
- Miniature Eccentric Rotating Mass (ERM) vibration motors
- Piezoelectric vibration, bending, or surface actuators
- Electromagnetic or voice-coil haptic actuators
- Hybrid actuation elements integrated with expandable or inflatable structures

Actuators may be selected based on compatibility with internal expansion systems, sealing requirements, and localized stimulus control.

Integration with Expansion and Compliance Systems

In certain embodiments, vibration or haptic actuation is integrated with one or more expandable or mechanically actuated components, including but not limited to:

- Annular inflatable cuffs or microbladder rings
- Radially expandable constraint bands
- Mechanically driven expansion elements
- Manual or powered pneumatic systems

Vibration may be applied concurrently with expansion, sequentially, or independently, allowing combined assessment of tissue compliance, contact dynamics, and sensory response.

Actuator Placement and Distribution

One or more actuators may be positioned within the device, including but not limited to:

- Circumferential placement surrounding the insertable body
- Longitudinal placement along the shaft or measurement zones
- Zoned placement aligned with pressure, contact, or distance sensors
- Embedded placement within expandable cuffs or compliant layers

Mechanical isolation structures may be employed to reduce interference with sensing elements.

Stimulus Characteristics

The vibration or haptic output may be modulated across one or more parameters, including:

- Frequency or resonance characteristics
- Amplitude or displacement
- Pulsed, continuous, ramped, or patterned waveforms
- Spatial sequencing or phase-shifted actuation across multiple zones

Stimulus profiles may be fixed, adjustable, or dynamically controlled.

Functional Roles

The vibration or haptic subsystem may serve one or more functional purposes, including:

- Controlled arousal or sensory engagement
- Neuromuscular or tissue response activation
- Perceptual confirmation of contact or engagement
- Augmentation of compliance or elasticity assessment
- Biofeedback correlated with pressure, expansion, or contact metrics

Closed-Loop and Sensor-Coupled Operation

In certain embodiments, vibration or haptic actuation is controlled in a closed-loop manner using real-time sensor input, including pressure transducers, circumferential contact sensors, capacitive sensing, distance sensors, temperature sensors, or inertial measurements. This enables adaptive stimulus delivery based on detected tissue response or device state.

Safety and Control

The device may incorporate safety mechanisms such as maximum expansion limits, vibration intensity limits, duty-cycle constraints, automatic shutoff conditions, user override controls, and context-aware inhibition to ensure safe and repeatable operation during intraluminal use.